# Inner vs. Inter City Transmission in the Hadith Social Network [Figures]

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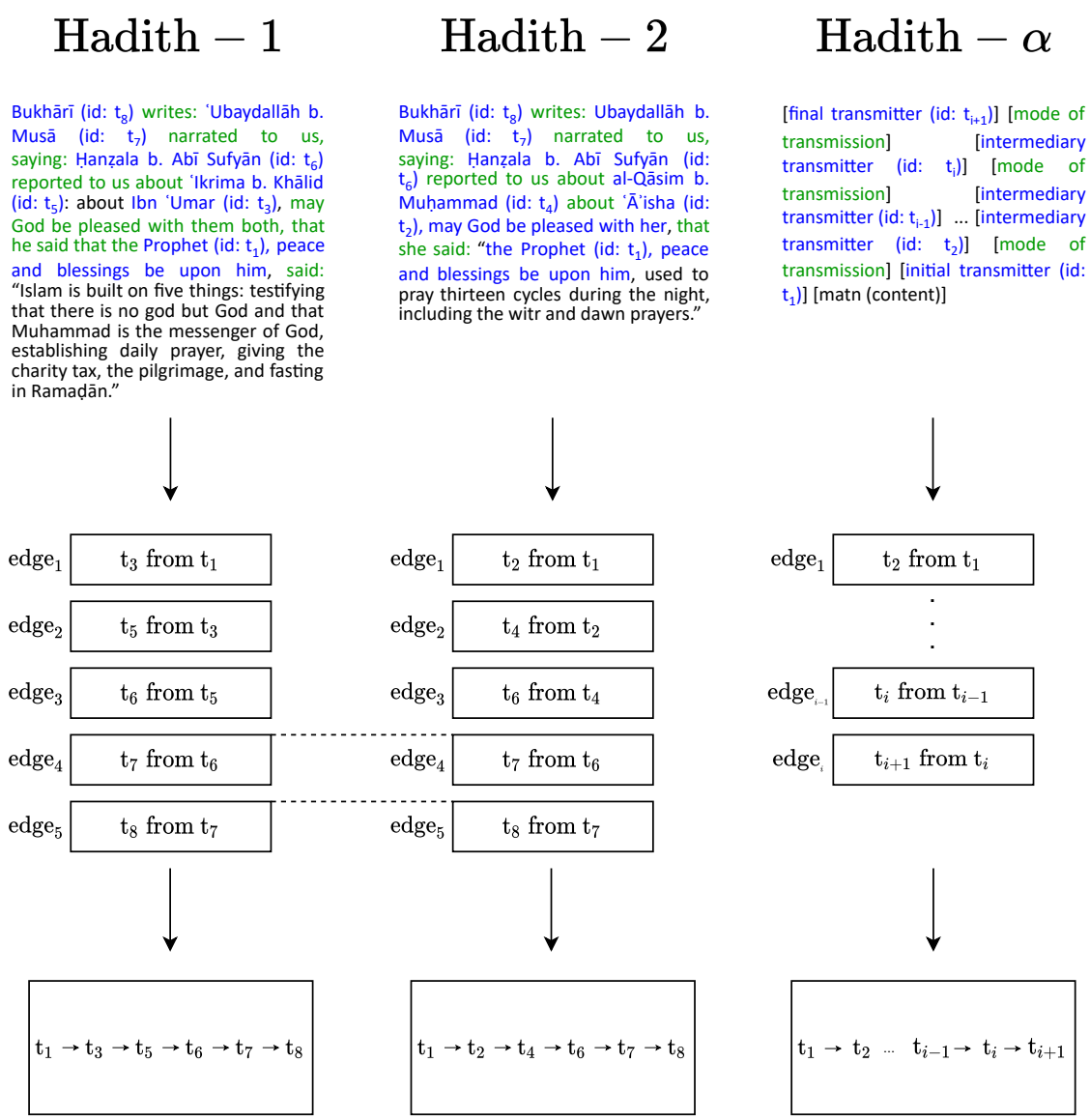
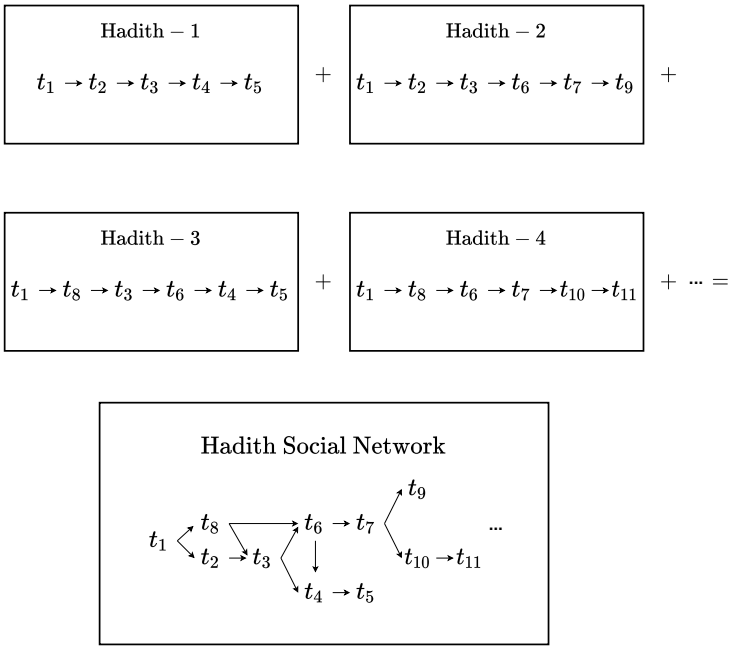
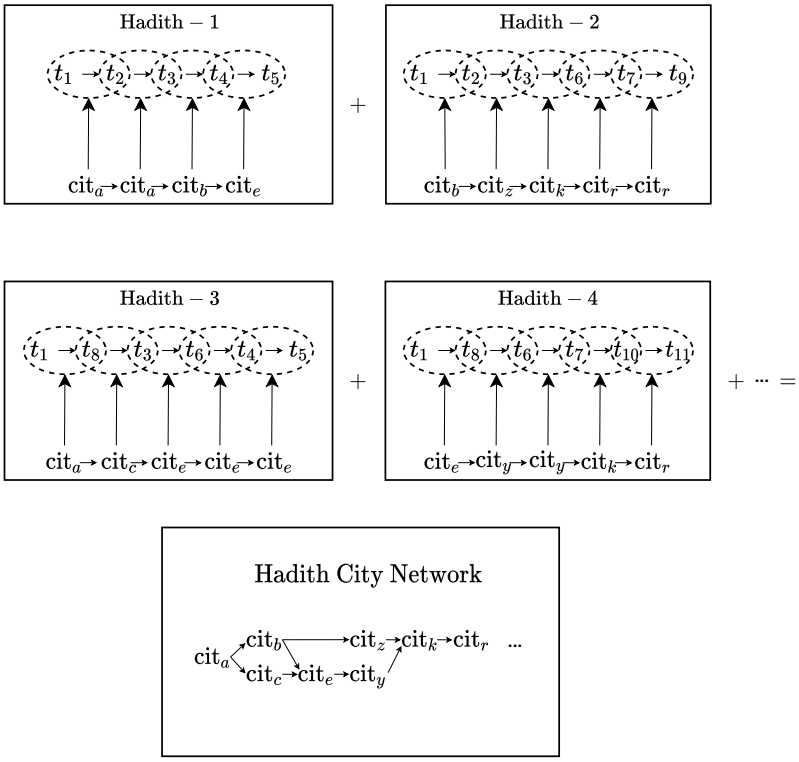
# 1 Hadith to Edges



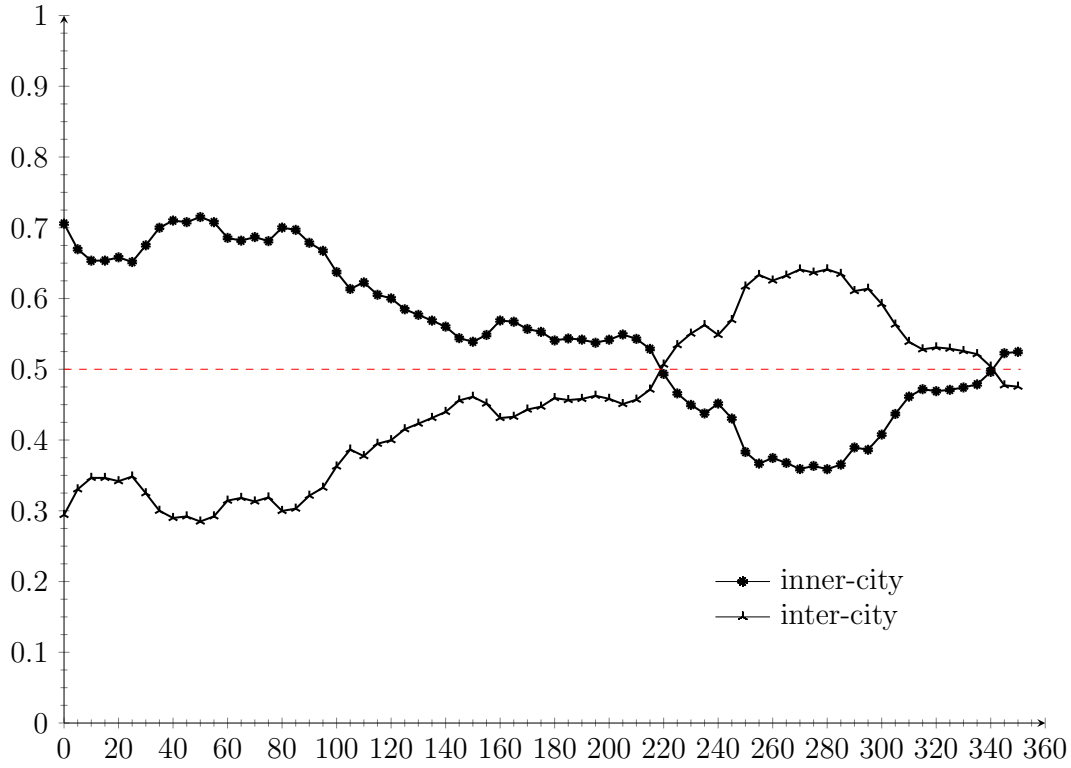
## 2 Edges to HSN



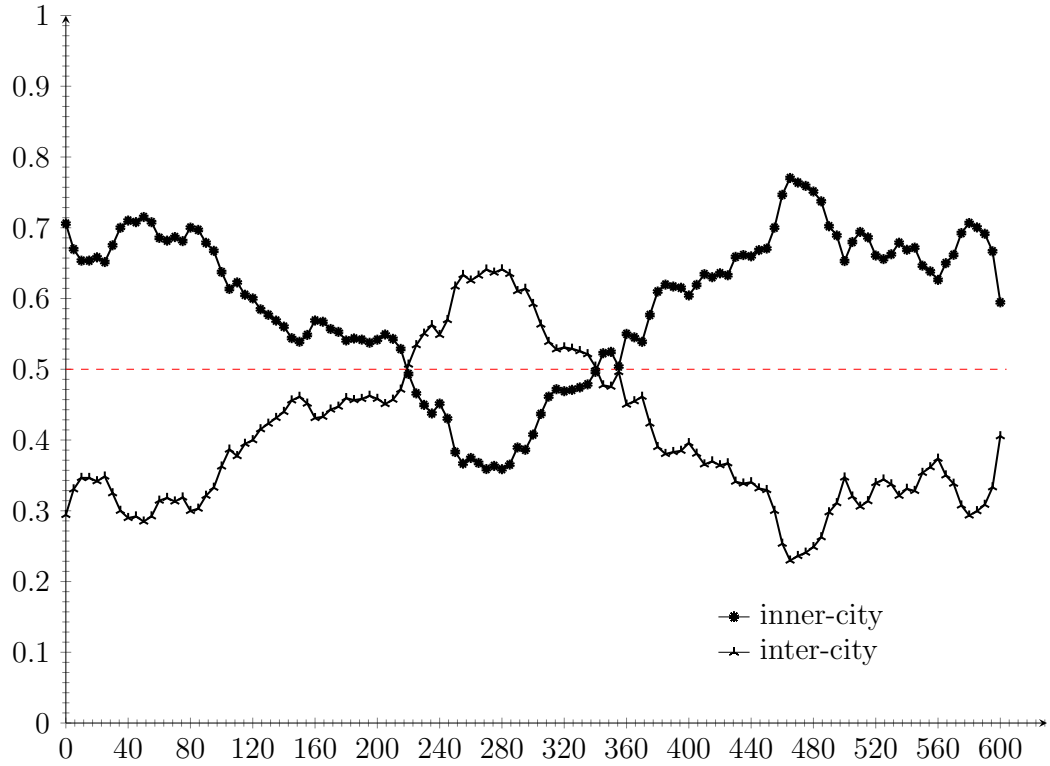
# 3 Edges to City Network



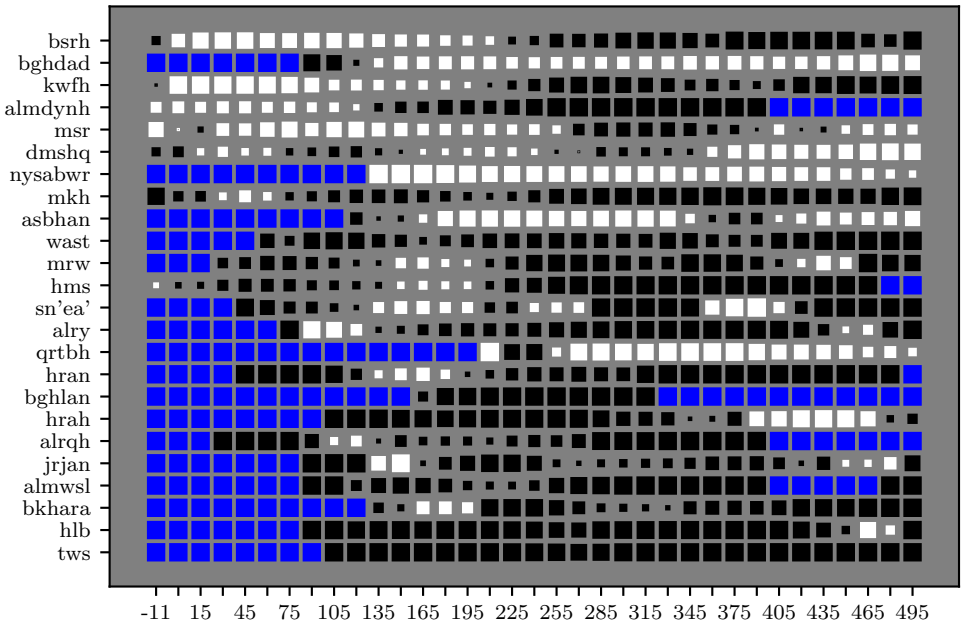
# 4 Bird's-eye View First 350 Years



# 5 Bird's-eye View First 600 Years



# 6 City Analysis



# 7 Transmitter Analysis (Inner)

| year span | | transmitter | inner | inter | ratio | transmitter | inner | inter | ratio |
|---|---|---|---|---|---|---|---|---|---|
| -11 | 0 | 4049 | 2543 | 1 | 0.9996 | 3436 | 8849 | 4573 | 0.6593 |
| 0 | 15 | 4945 | 260 | 2 | 0.9924 | 4945 | 260 | 2 | 0.9924 |
| 15 | 30 | 5913 | 913 | 2 | 0.9978 | 5913 | 913 | 2 | 0.9978 |
| 30 | 45 | 5543 | 246 | 2 | 0.9919 | 5722 | 1021 | 777 | 0.568 |
| 45 | 60 | 4049 | 2543 | 1 | 0.9996 | 4396 | 5949 | 5 | 0.9992 |
| 60 | 75 | 4980 | 770 | 1 | 0.9987 | 4967 | 3665 | 728 | 0.8342 |
| 75 | 90 | 2069 | 2817 | 5 | 0.9979 | 2069 | 2817 | 5 | 0.9979 |
| 90 | 105 | 5739 | 659 | 5 | 0.9922 | 4903 | 1838 | 22 | 0.9877 |
| 105 | 120 | 7863 | 3500 | 17 | 0.9949 | 6458 | 3500 | 834 | 0.8075 |
| 120 | 135 | 6399 | 145 | 1 | 0.9932 | 7272 | 4993 | 3160 | 0.6124 |
| 135 | 150 | 6712 | 396 | 3 | 0.9925 | 3629 | 6202 | 1720 | 0.7828 |
| 150 | 165 | 7133 | 278 | 1 | 0.9929 | 3436 | 8849 | 4573 | 0.6593 |
| 165 | 180 | 6437 | 288 | 3 | 0.9877 | 2492 | 8146 | 2338 | 0.777 |
| 180 | 195 | 4553 | 578 | 2 | 0.9953 | 6904 | 3376 | 67 | 0.9803 |
| 195 | 210 | 4244 | 295 | 1 | 0.9966 | 8160 | 6936 | 2066 | 0.7705 |
| 210 | 225 | 5938 | 360 | 1 | 0.9953 | 4533 | 6210 | 5175 | 0.5454 |
| 225 | 240 | 5435 | 594 | 1 | 0.9983 | 5049 | 6627 | 9838 | 0.4025 |
| 240 | 255 | 5219 | 129 | 1 | 0.9923 | 6879 | 6411 | 129 | 0.9802 |
| 255 | 270 | 12613 | 631 | 1 | 0.9968 | 8613 | 2244 | 125 | 0.947 |
| 270 | 285 | 17654 | 119 | 1 | 0.9905 | 12383 | 1437 | 430 | 0.7697 |
| 285 | 300 | 34407 | 105 | 1 | 0.9906 | 11603 | 1773 | 1421 | 0.5551 |
| 300 | 315 | 11548 | 231 | 1 | 0.9957 | 10025 | 2964 | 8106 | 0.2678 |
| 315 | 330 | 10596 | 476 | 1 | 0.9979 | 23051 | 2507 | 641 | 0.7962 |
| 330 | 345 | 10437 | 182 | 3 | 0.9837 | 27677 | 648 | 38 | 0.9435 |
| 345 | 360 | 22973 | 154 | 1 | 0.9918 | 28131 | 1515 | 24 | 0.9843 |
| 360 | 375 | 26320 | 547 | 2 | 0.9959 | 8994 | 3976 | 40 | 0.9899 |
| 375 | 390 | 22975 | 574 | 2 | 0.9955 | 24549 | 801 | 612 | 0.5668 |
| 390 | 405 | 18175 | 529 | 1 | 0.9977 | 26123 | 781 | 9 | 0.9884 |
| 405 | 420 | 41612 | 81 | 1 | 0.9878 | 13204 | 806 | 12 | 0.985 |
| 420 | 435 | 46431 | 91 | 1 | 0.9892 | 14773 | 744 | 51 | 0.9354 |
| 435 | 450 | 29523 | 148 | 1 | 0.9933 | 28387 | 207 | 11 | 0.9488 |
| 450 | 465 | 10610 | 210 | 1 | 0.995 | 29539 | 494 | 5 | 0.9898 |
| 465 | 480 | 46135 | 351 | 1 | 0.9972 | 21277 | 908 | 250 | 0.784 |
| 480 | 495 | 9601 | 312 | 3 | 0.9904 | 24779 | 528 | 38 | 0.9329 |
| 495 | 510 | 46384 | 95 | 1 | 0.9896 | 27317 | 325 | 10 | 0.9687 |

# 8 Transmitter Analysis (Inter)

| year span | | transmitter | inner | inter | ratio | transmitter | inner | inter | ratio |
|---|---|---|---|---|---|---|---|---|---|
| -11 | 0 | 30371 | 1 | 768 | 0.9987 | 3795 | 6140 | 13990 | 0.695 |
| 0 | 15 | 6618 | 1 | 3 | 0.75 | 6618 | 1 | 3 | 0.75 |
| 15 | 30 | 322 | 1 | 27 | 0.9643 | 7547 | 109 | 179 | 0.6212 |
| 30 | 45 | 3476 | 1 | 184 | 0.9946 | 5079 | 122 | 1750 | 0.9344 |
| 45 | 60 | 5672 | 1 | 304 | 0.9967 | 5021 | 2 | 485 | 0.9949 |
| 60 | 75 | 6259 | 1 | 81 | 0.9878 | 4883 | 2749 | 2806 | 0.5051 |
| 75 | 90 | 4622 | 1 | 127 | 0.9922 | 3929 | 14 | 780 | 0.9824 |
| 90 | 105 | 720 | 6 | 220 | 0.9692 | 2840 | 562 | 606 | 0.5186 |
| 105 | 120 | 4002 | 4 | 276 | 0.9858 | 5625 | 1261 | 1065 | 0.4579 |
| 120 | 135 | 4612 | 1 | 141 | 0.993 | 7272 | 4993 | 3160 | 0.3876 |
| 135 | 150 | 2622 | 1 | 123 | 0.992 | 3629 | 6202 | 1720 | 0.2172 |
| 150 | 165 | 8621 | 1 | 100 | 0.9902 | 3795 | 6140 | 13990 | 0.695 |
| 165 | 180 | 2201 | 1 | 165 | 0.994 | 2492 | 8146 | 2338 | 0.223 |
| 180 | 195 | 2450 | 1 | 127 | 0.9922 | 4716 | 189 | 3086 | 0.9421 |
| 195 | 210 | 3741 | 3 | 574 | 0.9944 | 3443 | 4433 | 13741 | 0.7561 |
| 210 | 225 | 1326 | 1 | 172 | 0.9942 | 4533 | 6210 | 5175 | 0.4546 |
| 225 | 240 | 3840 | 1 | 478 | 0.9979 | 5049 | 6627 | 9838 | 0.5975 |
| 240 | 255 | 5390 | 1 | 353 | 0.9972 | 488 | 3995 | 12611 | 0.7594 |
| 255 | 270 | 448 | 1 | 250 | 0.996 | 6817 | 663 | 3268 | 0.8313 |
| 270 | 285 | 30371 | 1 | 768 | 0.9987 | 28612 | 260 | 1415 | 0.8444 |
| 285 | 300 | 9844 | 1 | 419 | 0.9976 | 24433 | 29 | 3375 | 0.9914 |
| 300 | 315 | 25528 | 1 | 365 | 0.9973 | 10025 | 2964 | 8106 | 0.7322 |
| 315 | 330 | 9869 | 1 | 113 | 0.9912 | 22334 | 55 | 791 | 0.9345 |
| 330 | 345 | 30623 | 1 | 222 | 0.9955 | 10289 | 102 | 1213 | 0.9223 |
| 345 | 360 | 9866 | 1 | 712 | 0.9986 | 31063 | 422 | 3391 | 0.8891 |
| 360 | 375 | 28652 | 1 | 29 | 0.9673 | 22989 | 947 | 1387 | 0.594 |
| 375 | 390 | 17165 | 3 | 84 | 0.9643 | 28230 | 191 | 1492 | 0.8861 |
| 390 | 405 | 16463 | 3 | 303 | 0.9897 | 28679 | 83 | 527 | 0.8638 |
| 405 | 420 | 27629 | 6 | 748 | 0.9908 | 27629 | 6 | 748 | 0.9908 |
| 420 | 435 | 41400 | 1 | 90 | 0.9891 | 9927 | 147 | 952 | 0.8658 |
| 435 | 450 | 14991 | 1 | 309 | 0.9968 | 14991 | 1 | 309 | 0.9968 |
| 450 | 465 | 11867 | 1 | 322 | 0.9969 | 10034 | 489 | 543 | 0.5261 |
| 465 | 480 | 62375 | 1 | 176 | 0.9944 | 21277 | 908 | 250 | 0.216 |
| 480 | 495 | 46105 | 1 | 22 | 0.9578 | 18850 | 25 | 67 | 0.7287 |
| 495 | 510 | 63281 | 1 | 5 | 0.8453 | 27317 | 325 | 10 | 0.03131 |